Assessment Brief

Module title:	Data Visualisation
Module code:	COM7021
Assignment title:	COM7021 Assessment
Assignment format:	Report
Word/time limit:	3000
File type	.docx file
Percentage of final grade	This assignment is worth 100% of your final grade for this module.
Submission deadline	See module iLearn page for date of submission
Grade release	You will normally receive your provisional grade and feedback within 20 working days of the submission deadline

Useful terms:

Learning outcomes (LOs)	The skills and knowledge that you should be able to show in your work.
Marking Matrix	A set of rules or guidelines used to grade or assess work.

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Task summary:

This assignment is given as two tasks. Each task is to be answered individually, unless multiple components have been given.

- Task 1 has been designed to check your understanding of Data Visualisation methodologies for critical analysis, through a real-world mini–Data Visualisation investigation. **All of which will be conducted using python**.
- Task 2 is designed to demonstrate Data Visualisation reporting skills, comprehensively written in an easily understandable manner, based upon your Data investigation.

A clear, concise analysis for all Tasks is to be given within the submission, complimented with screenshot evidence of all processes and results. **You are to submit a single Word document for all tasks. The python code for your investigation is to be placed within the appendix so that it can be verified. Your student ID number must be clearly defined upon the uploaded file. All data investigations and visualisations will be conducted using Python**.

Assignment instructions:

For this assessment you must produce a data visualisation investigation (Task 1) and subsequent full report (Task 2).

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Task 1 – Data Visualisation in python

You have recently secured a position as a data visualisation creator at a Data Insights research facility. Your line manager has asked you to create appropriate visualisations using the **[5442] EcoMoveMobility.xlsx** dataset.

EcoMove City Transport Authority operates public transport and shared mobility services across several European cities. The dataset includes information such as date, city, country, transport mode, route type, weather, event day, passengers, trips, delay minutes, on-time percentage, ticket revenue, operating cost, CO2 saved, energy consumption, customer rating, and accessibility complaints.

Your analysis should explore how sustainable urban mobility performance varies across time, cities, countries, transport modes, weather conditions, major event days, and route types. The organisation is particularly interested in identifying high-demand cities, weak-performing transport modes, delay patterns, accessibility issues, cost and revenue performance, customer satisfaction, and environmental impact.

Using Python, produce suitable visualisations and an interactive dashboard that presents a clear data-driven narrative for the target audience. Your work should include **at least six different visualisations** and **one interactive dashboard**. The dashboard may be developed using Plotly Dash, Streamlit, Panel, Bokeh, or another suitable Python-based visualisation framework. The visualisations should not repeat the same chart type unnecessarily and should demonstrate a range of techniques, such as:

1. A time-series visualisation to show mobility trends over time
2. A comparison chart to compare cities, countries, route types, or transport modes
3. A relationship-based visualisation, such as a scatter plot or correlation heatmap
4. A distribution-based visualisation, such as a histogram, box plot, or violin plot
5. A performance indicator visualisation showing measures such as revenue, cost, delays, satisfaction, demand, or CO2 saved
6. A dashboard visualisation combining multiple views with interactive filters

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Justify why each visualisation was selected and explain how it supports the intended audience and decision-making purpose. The dashboard should help senior managers make decisions about route planning, service reliability, sustainability performance, accessibility improvements, and investment priorities.

The full investigation is to be conducted in Python. Ensure your Python code is submitted as an appendix to the report.

(70 Total marks)

(LO's: 1 & 2)

(1500 Word equivalent for analysis)

(600 Words)

Task 2 – Data Visualisation report

Based on your visualisation investigation in Task 1, produce a written and visually supported report that critically reflects on your analysis of the **[5442] EcoMoveMobility.xlsx** dataset.

Your report should justify the selected visualisations and interactive dashboard, explaining how they help technical and non-technical audiences understand transport demand, delay patterns, city-level performance, cost and revenue, customer experience, accessibility concerns, and sustainability impact.

You should present the key findings, conclusions, and evidence-based recommendations. The report should also briefly evaluate the effectiveness of your dashboard and explain how data visualisation can support business, social, environmental, and public-sector decision-making.

(30 Total marks)

(LO's: 3 & 4)

(900 Words)

Essential assignment resources Data Set

The data set needed to complete your assignment is entitled **[5442] EcoMoveMobility.xlsx** . A link to the data set is provided.

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Learning outcomes (LO)

By completing this assessment, you will have shown and be assessed on **all** four of the learning outcomes:

1. Critically appraise a variety of data visualisation techniques in terms of psychology, cultural context, effectiveness, and target audience LO 1

2. Apply appropriate data visualisation techniques to analyse and present data LO 2

3. Demonstrate a critical understanding of how data visualisation can support technical and non-technical audiences in making business decisions. LO 3

4. Graduate Attribute:

Critically evaluate and further contextualise understanding of subject area with the ability to link the discipline(s) to local, national and global issues, research and pursue evidence-based arguments within the discipline using an extensive range of academic and professional body resources. Show clear ability to engage with different traditions of thought, and the ability to apply their knowledge in practice including in multidisciplinary or multi-professional context LO4

Your marker will use a rubric/ marking matrix to grade your work, and you can access this by clicking on the submission portal.

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